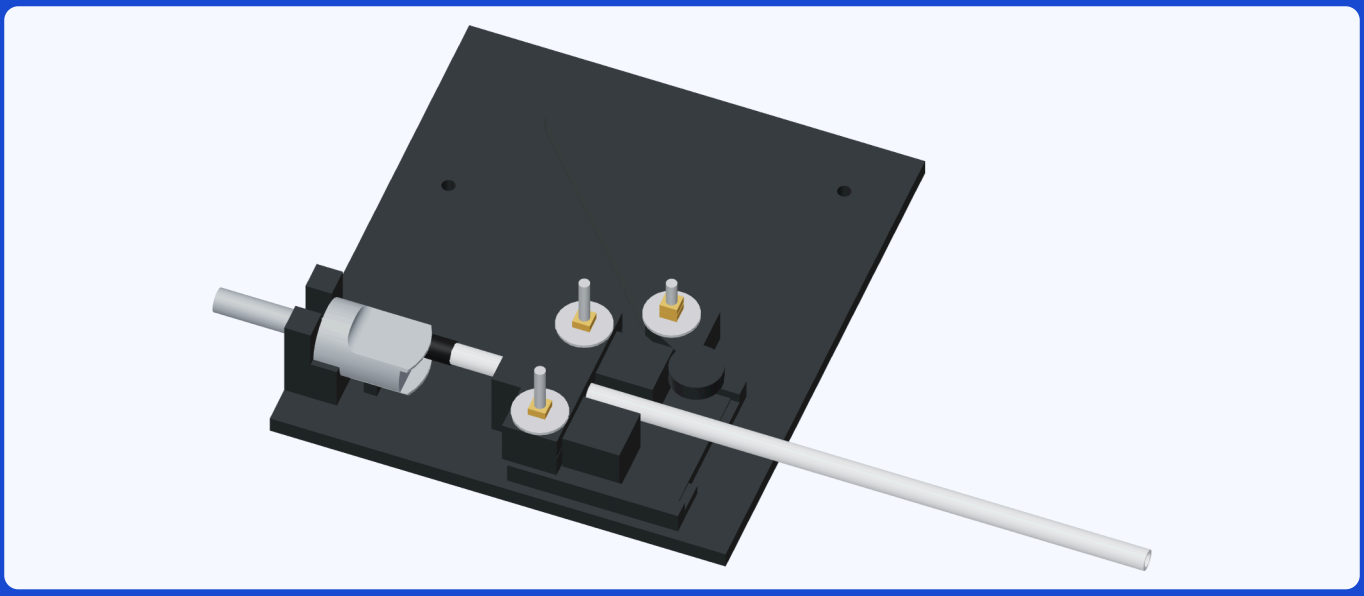
HOME SODA MACHINE

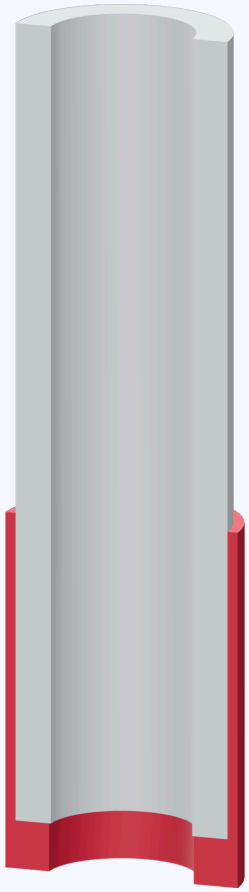
Tube insertion jig

Build the T handle and bench lever press. Fit the TPU thimble onto the soda tube, then press the sleeved tube into the bare Touch-Flo.



The tube arrives sleeved

Two separate operations: stretch the thimble onto the tube outside the donor, then insert that sleeved end into the metal port.



The square-cut tube end rests on the thimble's annular cap. The cap hole remains open into the tube bore.

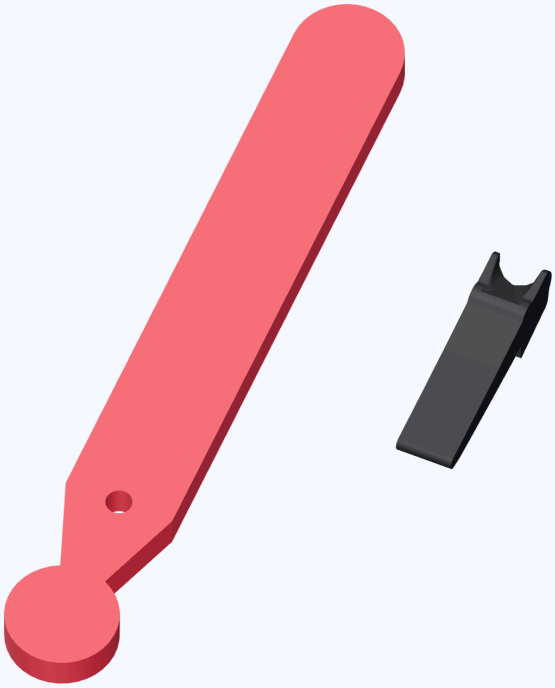
1. Attach the accepted printed faucet lever to the bare Touch-Flo. Leave it **down at rest, unpressed**.
2. Use the T handle to hold the tube while fitting the TPU thimble evenly around its end.
3. Place the bare Touch-Flo in the nest and press the **sleeved tube** into its empty top port.
4. Release the fixture. The faucet lever and sleeved tube remain attached when the donor enters the faucet shell.

THE THIMBLE GOES ONTO THE TUBE FIRST

TPU slides against the metal far more easily than against LLDPE. Putting the thimble into the donor first can bunch it under the entering tube.

Two levers, two jobs

The long **press lever** provides insertion force. The small **faucet lever** operates the valve and stays attached throughout the operation.



The press lever belongs to the bench jig. The faucet lever is shown separately for identification and stays attached to the donor during insertion.

IN A PICTURE	MEANS
Coral	The part or fastener this step adds or moves. Existing printed structure stays black.
Brass	A threaded metal nut.
Blue	An alignment tool or reference mark.
Grey metal	The Touch-Flo or a fitted metal fastener.

Fasteners float off their own holes along the direction they enter. The picture identifies the joint; the stated size identifies the fastener. **Do not measure a picture.**

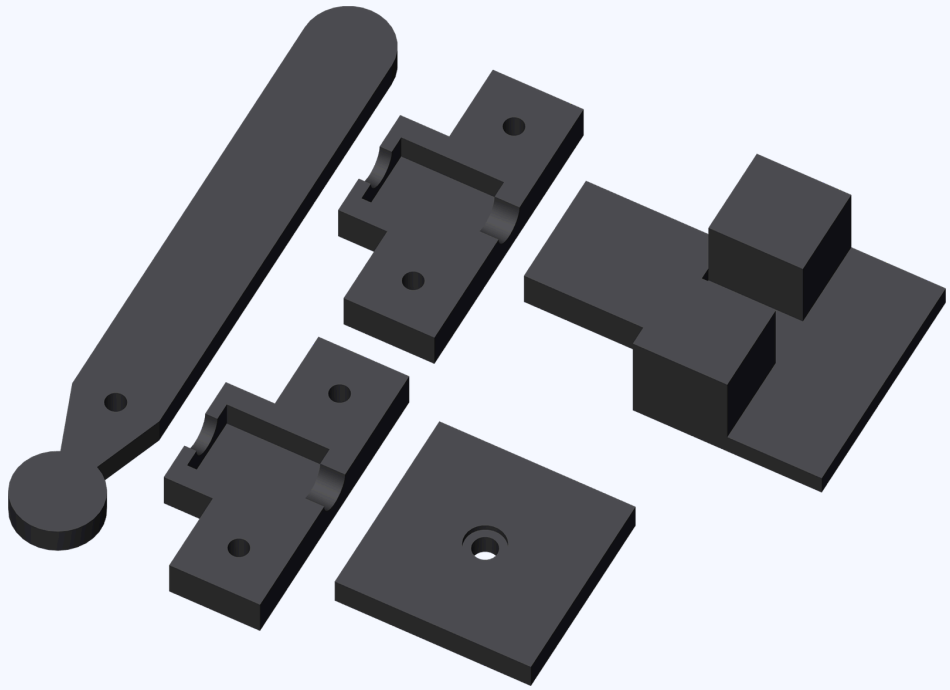
FINISHED WHEN

A box like this describes the observable result of the action on that page.

PRINT · OPERATION

One base, one parts plate

Open the prepared PET-GF projects for H2C / Mark1 with its left 0.4 mm nozzle. The manufacturing STLs already carry their bed orientation.



B ram, C press lever, D and E jaws, and G cap-support puck in their flat print orientations. Base A prints on its own plate.

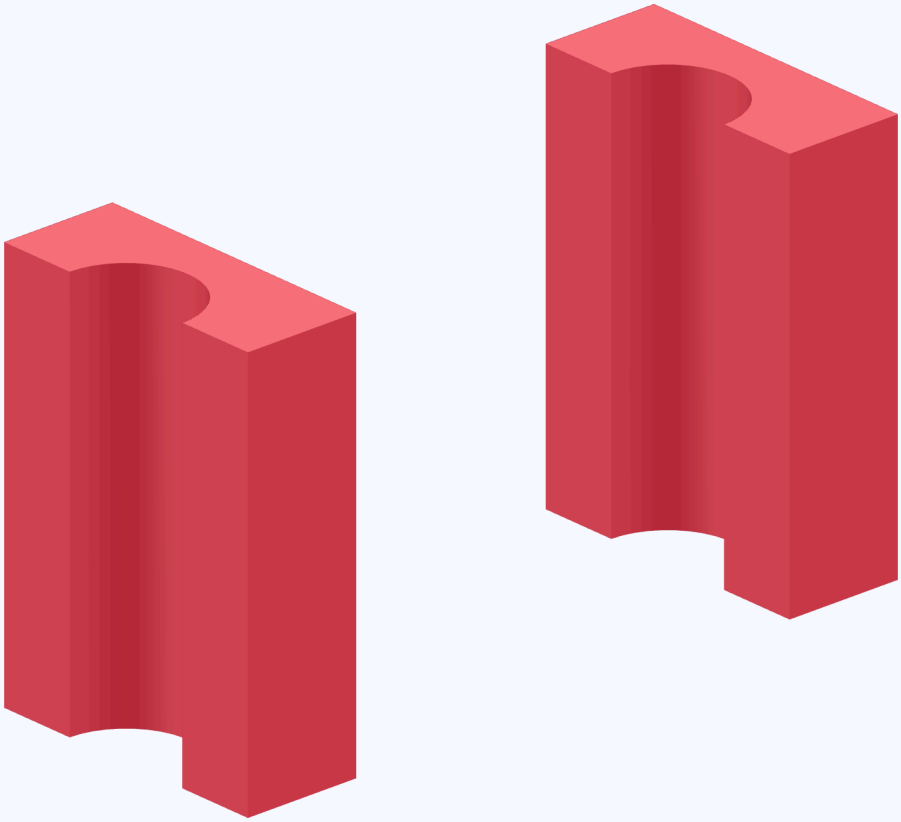
PART	QTY	ORIENTATION
A Base with nest, rails and stop	1	Broad bottom down.
B Ram	1	Flat feet down; open fork up.
C Press lever	1	Broad flat face down.
D / E Lower / upper jaws	1 each	Flat backs down; liner pockets up.
G Cap-support puck	1	Recess up.

`tube-insertion-rigid-base-h2c.3mf`
`tube-insertion-rigid-parts-h2c.3mf`

Use the saved project orientations and support settings. Their matching `.gcode.3mf` archives are the prepared native slices.

Print the finished grip shape

Each liner is a thick block with a smooth half-round channel. The two finished channels meet directly around the tube.



Two copies of F, already shaped to fit the tube. Each flat backing seats directly in a rigid jaw pocket.

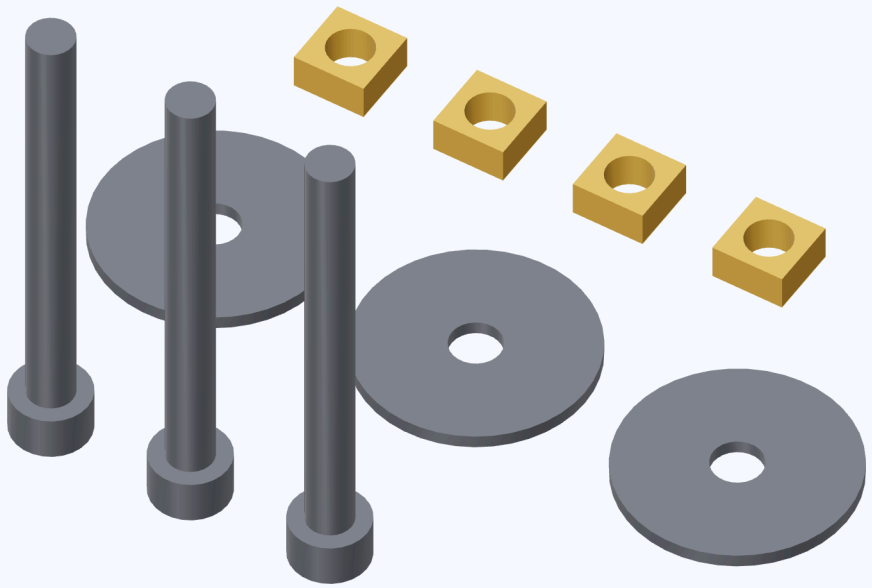
ITEM	PRINT
Grip liner	2 copies, TPU 85A. Use the prepared manufacturing STL orientation and saved project.
Print pose	Tube axis vertical. Each block prints 28 mm high on an 8 × 20 mm footprint; its minimum solid backing is 3.375 mm.
Saved project	Mark2 right 0.6 mm nozzle. `tube-insertion-tpu-85a-mark2.3mf`
Before assembly	Remove strings from the smooth contact channels and backing edges. Preserve the full backing and round gripping surface.

THE GRIP LINER AND WATER SEAL ARE DIFFERENT PARTS

These thick liners belong to the removable tool. The thin TPU 85A thimble belongs to the faucet's wet path and retains its existing geometry and print recipe.

Three bolts, four nuts

The T handle uses two bolted jaws. The press lever uses the third bolt and a pair of jammed square nuts.



Three M5 × 50 socket-head screws, three large washers and four square nuts.

HARDWARE	QTY	WHERE
M5 × 50 socket-head screw	3	2 jaw screws, 1 press pivot.
M5 × 0.8 square nut, 8 mm across flats × 4 mm thick	4	2 jaw nuts, 2 pivot jam nuts.
M5 fender washer, Ø25 mm × 1.5 mm, Ø5.5 mm bore	3	2 jaw load spreaders, 1 pivot thrust washer.

The identified bolts, square nuts and washers are already stocked. The fixture uses no heat-set inserts.

What you need at hand

The fixture bolts together. Its stop setup uses the actual donor's port and two marks on the tube.

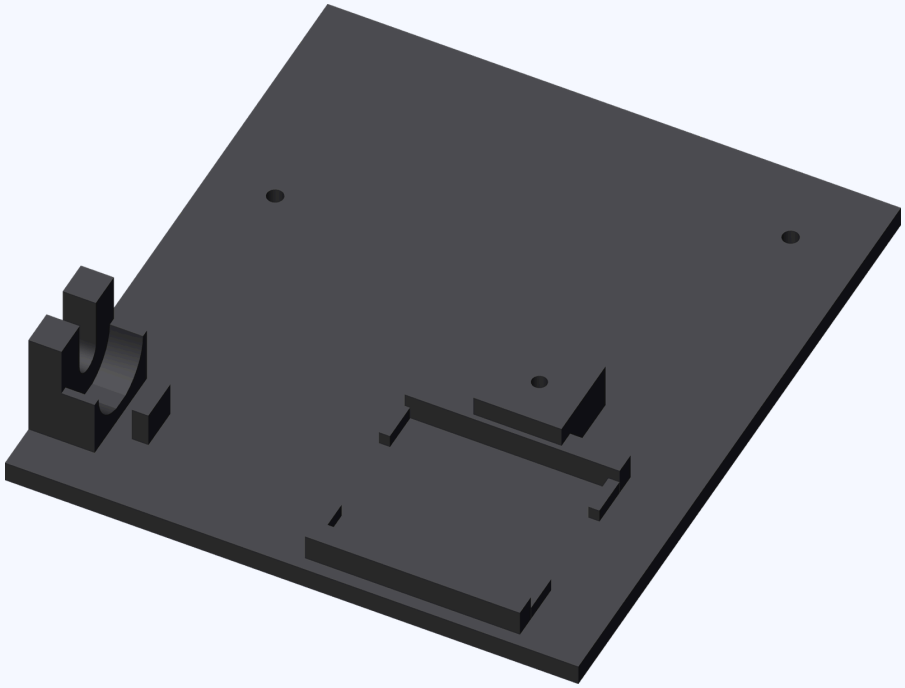
TOOL OR WORKPIECE	FOR
4 mm hex key	The M5 socket-head fixture screws and press pivot.
Two small wrenches or pliers	Hold the 8 mm square nuts while setting and jamming the two pivot nuts.
Digital caliper with depth blade	Measure the actual donor's annular port floor and locate the two tube marks.
Tube cutter	A square, clean cut on the 3/8-inch OD LLDPE soda tube.
Fine marker	A port-lip seat witness and a clamp-front position mark, measured from the bare tube's square-cut end.
Firm bench	Support the base. The opposite hand holds its fixed palm area during normal use.
Existing bench clamp or mounting bolts, if needed	Hold the base stationary if the palm hold cannot do so. The base provides two open mounting holes.
Plastic scraper, small cutters	Remove strings and print remnants from rail faces, liner pockets and gripping channels.
Bare Touch-Flo with accepted faucet lever	The workpiece. Its top port is empty when the sleeved tube approaches.
3/8-inch OD LLDPE soda tube	The faucet's finish-matched tube, prepared to the faucet assembly guide's length.
Fresh TPU 85A thimble	The existing faucet water seal, fitted onto the tube outside the donor.

HOLD THE BASE STATIONARY

The opposite hand holds the fixed palm area, away from the ram and press lever. If the base moves, use an existing clamp or mounting bolts before continuing.

Clean the working surfaces

Base A carries the donor nest, thrust wall, ram guides, stop and pivot pedestal as one print.



The base on its broad bottom. The donor shank passes through the open notch at the left; the opposite side provides the fixed palm area.

1. Remove the print remnants and any supports generated by the prepared project. Clear the underside pivot-head seat completely.
2. Remove strings and raised edges from the nest, rail faces, stop face, ram feet and nut/washer bearing faces.
3. Keep the thrust wall's annular metal-contact face and both rail faces flat. Preserve their printed bearing surfaces.

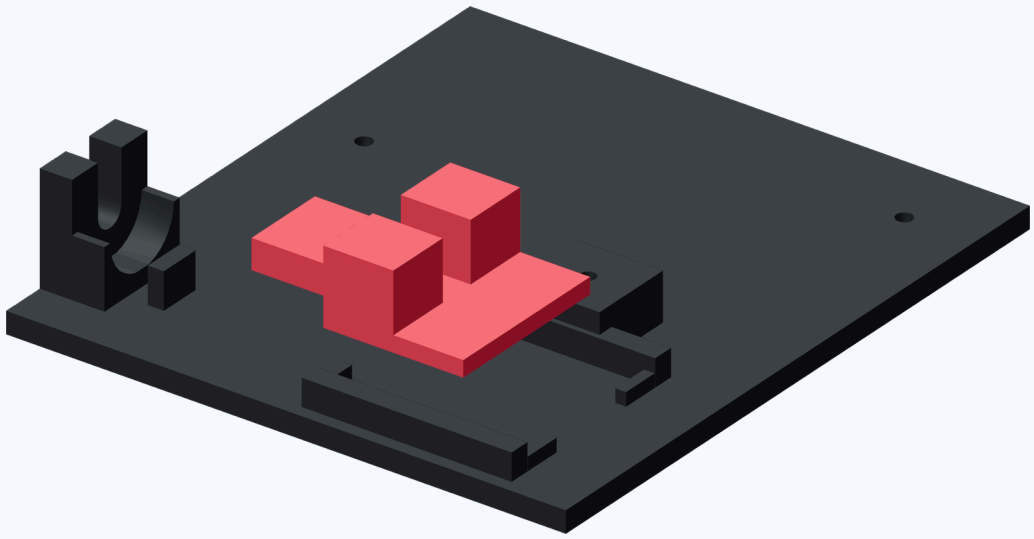
FINISHED WHEN

The ram path is clear, the donor nest is clean and the pivot screw can enter its underside seat.

BUILD · OPERATION

Drop B between the rails

The ram is open above the tube and at both ends. Its two feet slide on the base.



The ram poised above its guide channel. The fork faces the donor and its open top leaves the free tube accessible.

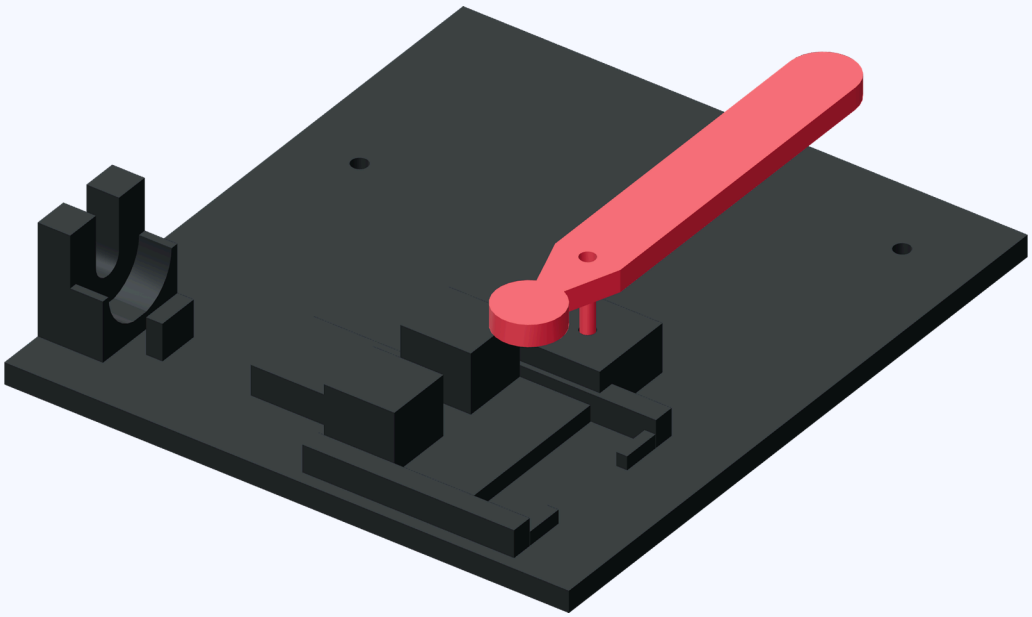
1. Set base A flat on the bench. Place ram B between the two rails with its feet down and fork up.
2. Slide the ram toward the donor until both guide feet meet the paired front stops.
3. Slide it back about 20 mm, then return it to the stop by hand. Keep it square between the rails.

FINISHED WHEN

The ram slides freely throughout the working stroke and both guide feet meet the front stops together. Remove a string or raised print edge if it catches.

Put C on its pivot

The short rounded nose drives the ram. The long handle stays on the opposite side of the free tube.



Press lever C poised above its pedestal; the M5 × 50 screw enters upward through the base.

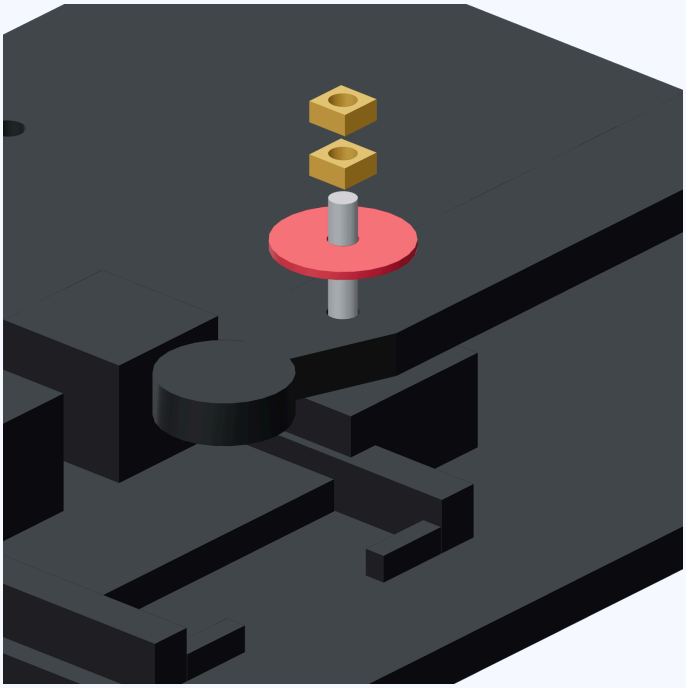
1. Pass one M5 × 50 socket-head screw up through the pivot hole from the underside. Its head sits in the base's accessible head recess.
2. Lower press lever C over the screw, flat on the pedestal, with its short nose toward the ram and long handle over the broad rear area.
3. Keep the ram at the forward stop while positioning the short nose against its rear drive face.

FINISHED WHEN

The screw passes through the lever's pivot bore and the lever is oriented with its short nose at the ram.

Retain a freely moving pivot

A large washer spreads the pivot load. Two square nuts hold the setting without clamping the printed lever tight.



One Ø25 mm washer and two M5 square nuts poised above the press-lever pivot.

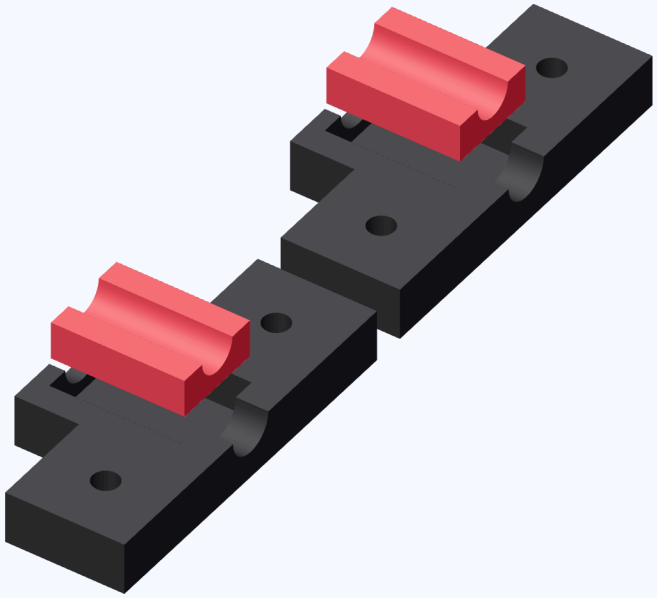
1. Place one Ø25 mm washer over the pivot screw and onto the lever.
2. Thread on one M5 square nut. Set it so the lever is retained with slight endplay and swings freely.
3. Thread on the second square nut. Hold the first nut at that setting while jamming the second nut against it with the two small wrenches or pliers.
4. Swing the lever through the stroke by hand. The nose should drive the square ram without binding or lifting it.

FINISHED WHEN

The two nuts remain jammed together, the lever stays retained and it swings without pinching at the pivot.

Seat one F in each jaw

The same thick liner fits each jaw. Its flat backing goes into the pocket and its smooth channel faces the tube.



D and E with their pockets facing up on the bench; one F liner poised over each pocket.

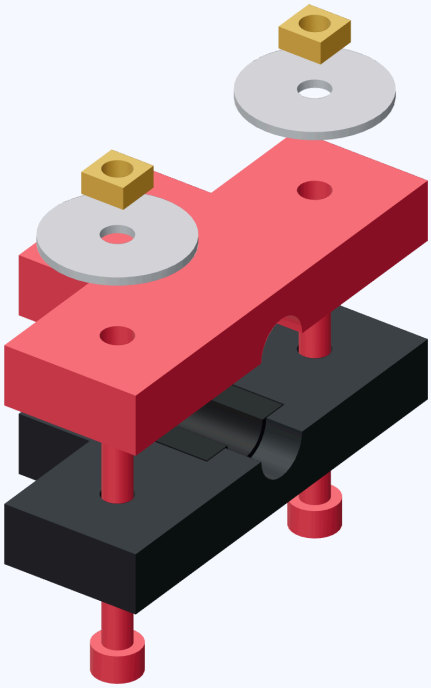
1. Lay lower jaw D and upper jaw E pocket-up on the bench.
2. Press one F liner fully into each pocket, flat backing against the pocket floor. Its half-round channel remains exposed.
3. Check each backing and end edge. Both liners must lie flat, without a trapped string or raised corner.

FINISHED WHEN

Both liners are seated flat. Their channels will face one another when the jaws close.

Bolt the two jaws together

The wide rear wings provide a hand grip and the shoulders the ram pushes. The narrow front faces the sleeved tube end.



Lower jaw D, upper jaw E, two M5 × 50 screws, two Ø25 mm washers and two M5 square nuts.

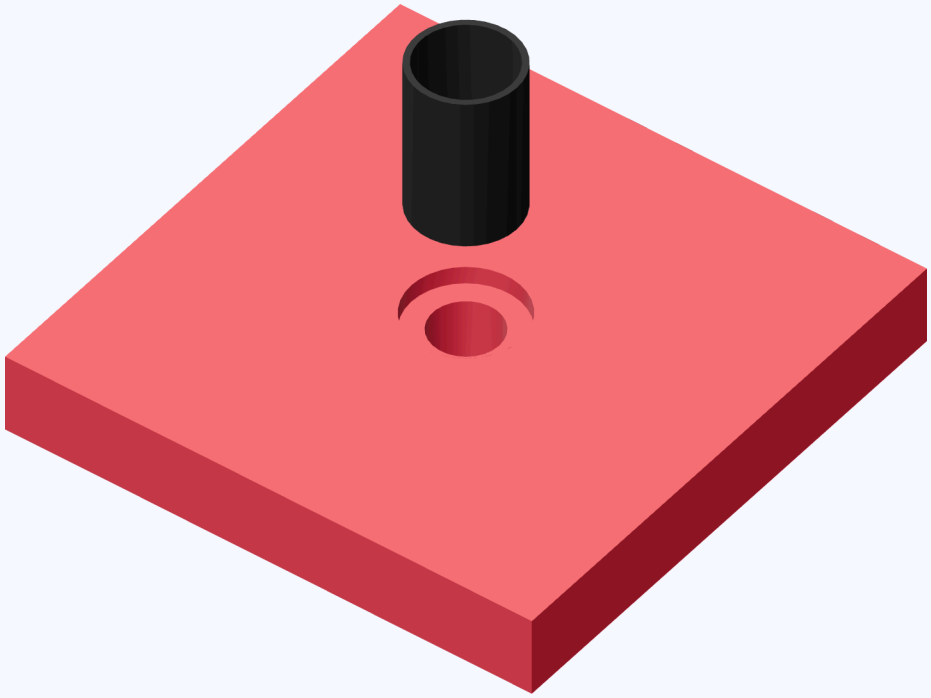
1. Pass one M5 × 50 screw upward through each of lower jaw D's two rear-wing holes.
2. Place upper jaw E over them with its liner channel facing the lower channel.
3. Put a Ø25 mm washer over each screw on the upper jaw, then thread on an M5 square nut.
4. Leave the two screws loose enough to open the jaws around a tube. When used, close the hard mating faces evenly; those faces limit the liner squeeze.

FINISHED WHEN

The jaw halves stay paired, open freely and close squarely. The upper washers sit fully on the rigid rear wings.

G supports the annular cap

The shallow recess locates the seal's cap while leaving its entire cylindrical wall accessible to your fingers.



The existing thimble over the puck's Ø11.2 mm, 1.8 mm-deep recess. The Ø6.8 mm through-hole leaves the seal's cap hole open.

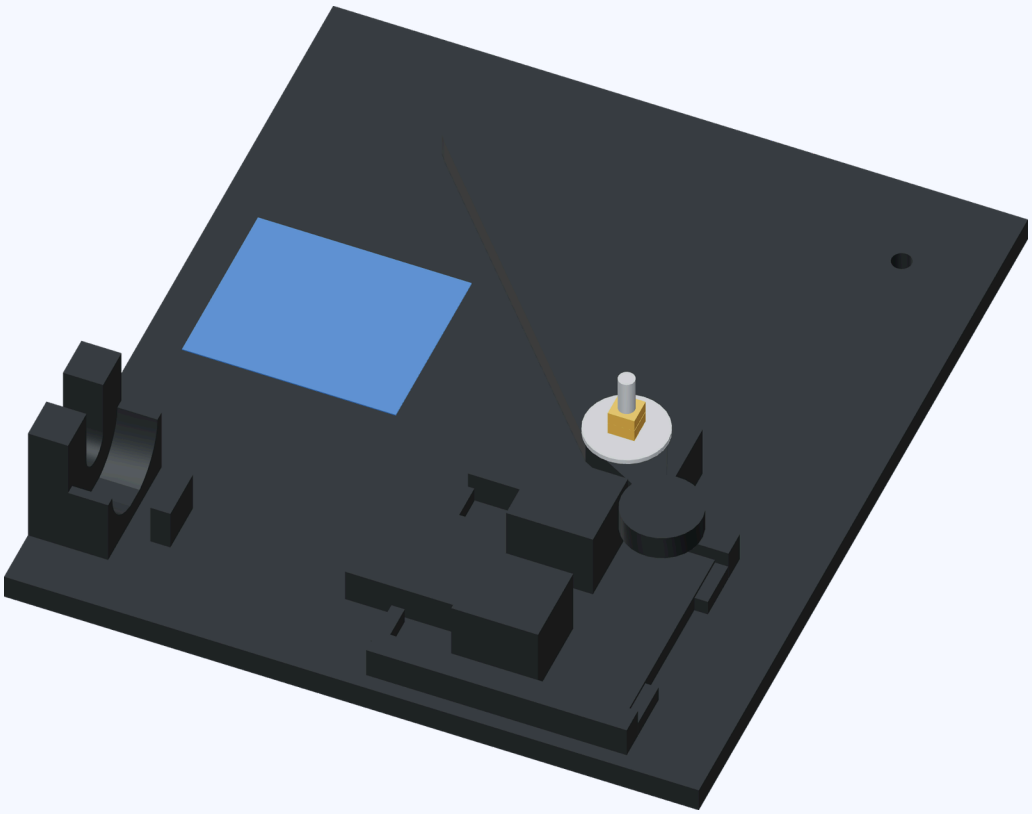
1. Clear the puck's through-hole and shallow recess of print remnants.
2. Set it recess-up on the bench. It receives the fresh thimble cap-down during sleeve fitting.
3. Keep the puck loose. It is a bench support, not a piece that enters the donor with the tube.

FINISHED WHEN

The recess is clean and the through-hole is open.

Use the fixed palm area

One hand holds the stationary base while the other moves the press lever. Keep the hand away from the moving ram and pivot.



Blue marks the fixed palm area beside the donor. It is a reference in the picture, not an added fixture part.

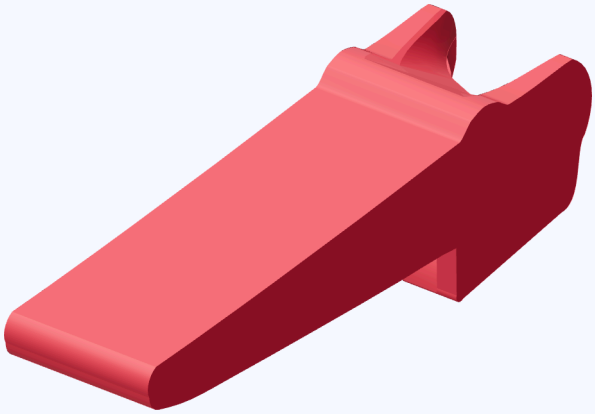
1. Set the base flat on a firm bench with room for the shank at the left and the free tube to the right.
2. Place the opposite hand on the fixed palm area beside the donor, on the side away from the faucet lever. Keep fingers outside the ram and press-lever paths.
3. If that hold cannot keep the base stationary, use an existing bench clamp or mounting bolts at the base's two open mounting holes.

FINISHED WHEN

The base remains flat and stationary while the empty press lever moves through its stroke.

Attach the faucet lever first

The bare Touch-Flo and its accepted faucet lever are already connected before the tube or the shell is fitted.



The accepted printed faucet lever, shown separately for identification. Its mounted rest position leaves the top water port accessible.

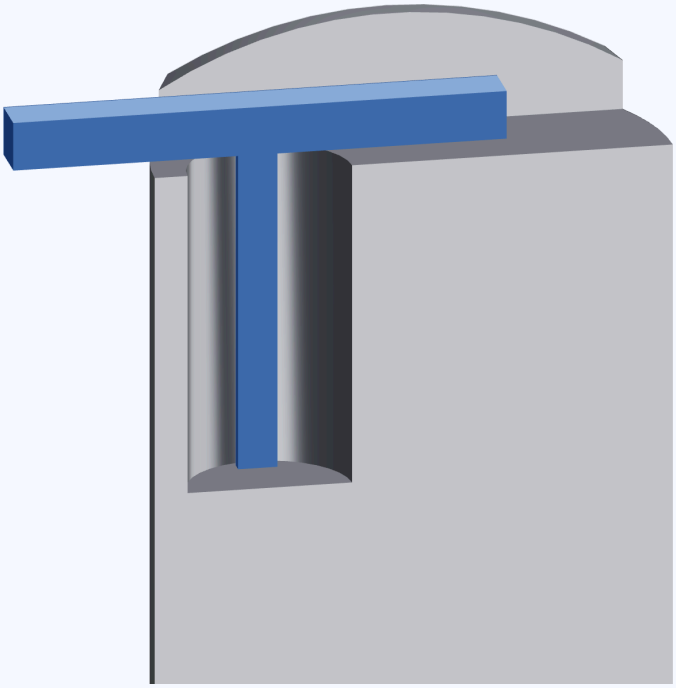
1. With the tube absent, lower the accepted faucet lever aft of its working position and slide it forward around the metal cylinder.
2. Leave the faucet lever **down at its unpressed rest**. Keep it attached for sleeve insertion and later shell assembly.
3. Confirm the donor's top port is empty. The thimble belongs on the tube before it enters this port.

FINISHED WHEN

The faucet lever is attached at unpressed rest and the top port is empty.

Measure the actual port depth

This reading sets where the tube grip sits. It allows the press to stop at the actual donor's seating depth without a prior manual full seating.



Caliper depth blade at the annular seating floor beside the central water opening. D is measured from the port mouth.

1. Use the existing caliper's depth blade. Rest its reference beam on the **port mouth below the raised arches**, clear of the plunger. An arch-top reading adds the wrong height.
2. Lower the blade gently beside the bore wall until it touches the **annular seating floor** that supports the thimble cap.
3. Read and record the depth as **D**, in millimetres. Check at the opposite side of the annular floor to confirm the same datum.
4. If the caliper cannot reach that datum, gently touch a clean 4 mm hex-key tip to the annular floor, mark the key at the port mouth, withdraw it and measure from the tip to the mark with the caliper.

MEASURE THE CAP SEAT

Keep the blade out of the central water opening and away from the valve plunger. The port's nominal 20 mm depth is not the fixture setup measurement.

Two marks from the tube end

Measure from the bare tube's square-cut end before the thimble hides it. The cap thickness is 2.1 mm.



Measure from the lower tube end shown. The lower mark is the port-lip witness; the upper mark locates the front of the jaws.

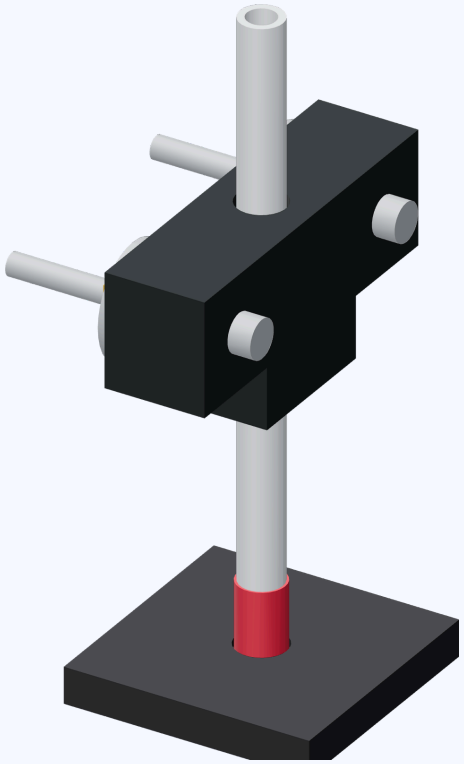
1. Confirm a square, clean tube end and an open bore. Use the tube length specified for the faucet finish.
2. Mark a fine line **D - 2.1 mm** from that end. This is the **port-lip seat witness**.
3. Mark a second line **D + 16.9 mm** from the same end. This is the **clamp-front mark**.
4. Keep the two lines distinguishable. The clamp-front line sits 19 mm farther up the tube than the port-lip witness.

FOR EXAMPLE ONLY

If the actual D reading is 20.0 mm, the lines lie 17.9 mm and 36.9 mm from the tube end. Use **your measured D**.

Work the thimble onto the tube

Use the T handle for grip and the cap-support puck for reaction. Fit the seal evenly outside the donor.



The puck supports the annular cap. The T handle holds the tube while the thimble wall advances evenly around it.

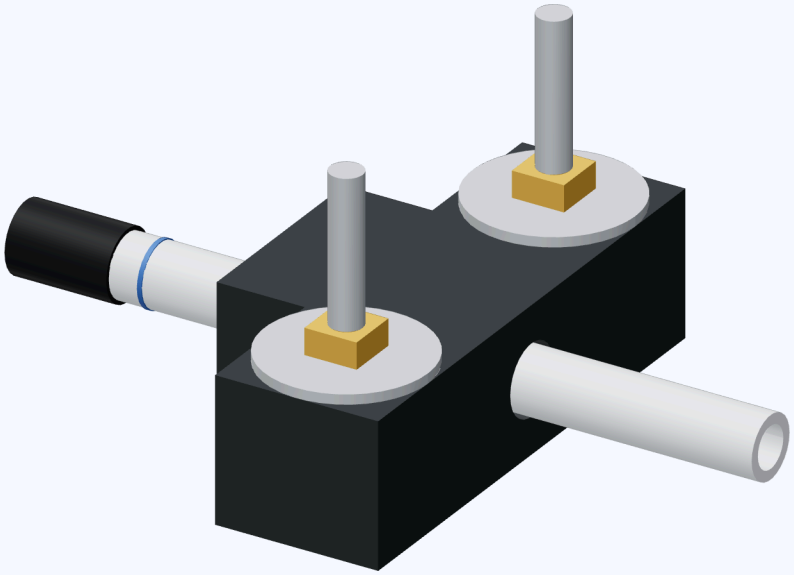
1. Grip the tube with the T handle far enough from the end to leave the entire thimble and both marks accessible.
2. Set the fresh TPU 85A thimble cap-down in the puck's shallow recess. Its open end faces the tube.
3. Center the square-cut tube end at the opening. Work the sleeve rim over it evenly around the circumference while advancing the tube.
4. Press through the T handle in small increments while supporting and stretching the rim evenly. Keep the sleeve straight until the tube end contacts the annular cap.
5. Lift the sleeved tube off the cap-support puck. The puck stays on the bench for the donor insertion operation.

FINISHED WHEN

The sleeve wall is even, with no fold or roll, and the square-cut tube end rests on the cap. The cap hole remains open into the tube bore.

Move the grip to its mark

The grip pushes the tube. The thimble and port remain free of the jaws.



The jaw front faces the sleeved end and aligns with the D + 16.9 mm clamp-front mark.

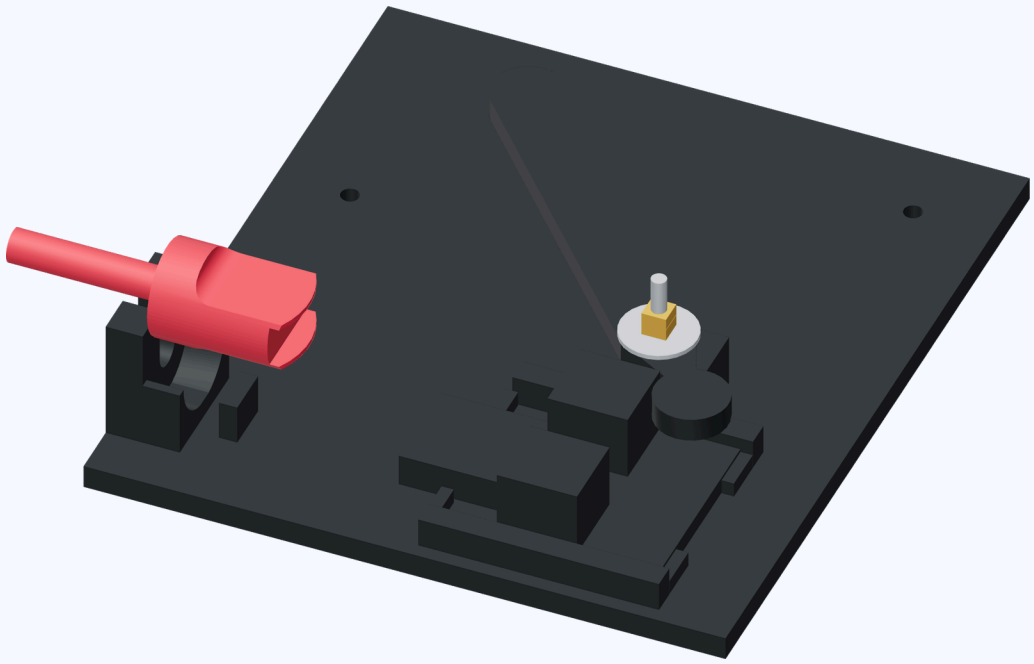
1. Loosen the two jaw screws enough to open the grip and relocate it without dragging a closed liner along the tube.
2. Place the two liners around the bare LLDPE with the **front edge of both jaws** at the clamp-front mark.
3. Close the rigid jaw shoulders evenly using the two M5 screws. The rigid closure shoulders limit compression; do not add a spacer or extra squeeze.
4. Confirm both marks are where intended and the tube end remains fully seated on the thimble's cap.

THE TUBE STAYS ROUND

Use the designed jaw closure. If the grip slips during insertion, or the tube stays flattened after release, stop and inspect the liner fit and jaw closure.

Place the donor in the nest

The horizontal fixture supports the robust cylindrical metal base. The attached faucet lever remains in the open front-side clearance.



The shank exits left and the water port faces the tube. The attached faucet lever is omitted from this CAD view; keep it down at unpressed rest on the open front side.

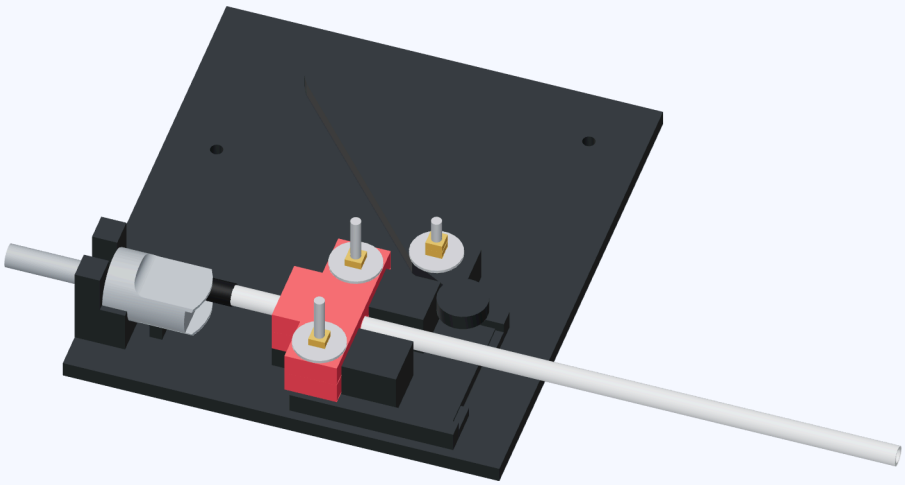
1. Lay the bare Touch-Flo in the nest with its cylindrical base in the saddle and its flat bottom annulus against the thrust wall.
2. Let its threaded shank pass through the open notch. Rest the flat metal column on the clocking pad so the empty top port aligns with the tube axis.
3. Keep the attached faucet lever **down at unpressed rest** in the open side facing you. The fixture must touch neither the lever nor the valve plunger.

FINISHED WHEN

The metal body rests against the thrust wall, the shank is clear and the unpressed faucet lever has open clearance.

Engage the two grip shoulders

The press provides up to 20 mm of guided travel. Use the T handle for any starting engagement needed by a deeper measured port.



A nominal 20 mm setup: both grip shoulders engage the ram, with the lower jaw on the full shelf. The attached faucet lever is omitted; keep it down at unpressed rest on the open front side.

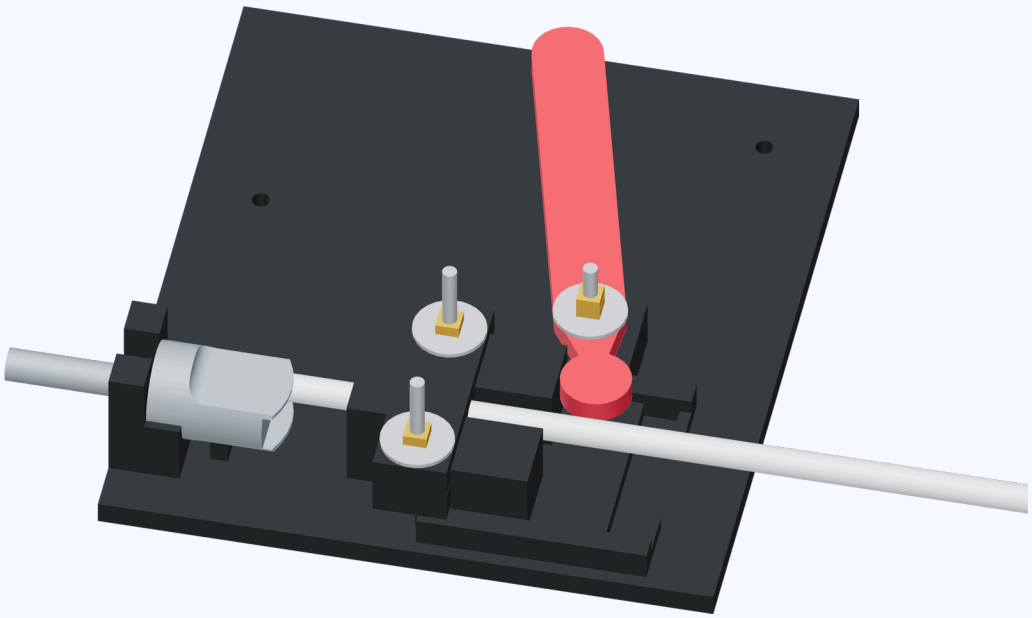
1. Move the press lever back and draw the ram fully away from the donor. Lay the tube through its open top and align the sleeved cap with the port.
2. For **D > 20 mm**, hold the base and use the T handle to start the sleeve **D - 20 mm** into the metal port. Stop when the grip's rear shoulders can reach the retracted ram.
3. For **D ≤ 20 mm**, keep the cap at the mouth and advance the unloaded ram **20 - D mm** until it reaches both grip shoulders. The loaded travel is D.
4. Set lower jaw D on the ram's full support shelf and engage both rigid rear shoulders against the two ram faces. Keep the tube coaxial with the port.
5. Confirm the grip front is still at its clamp-front mark and the thimble is evenly fitted onto the tube.

FINISHED WHEN

The shelf supports the lower jaw, both shoulders share the ram contact and the tube is straight through the open passage.

Move the press lever slowly

The short nose moves the guided ram; the long handle provides approximately 4:1 leverage. The faucet lever remains unpressed.



The press lever moves the grip toward the port; the free tube continues beyond the picture. The attached faucet lever is omitted; keep it down at unpressed rest on the open front side.

1. Hold the fixed palm area with the opposite hand. Keep the donor's annular base face against the thrust wall.
2. Move the long press lever slowly through the stroke. Watch the thimble enter straight and the grip remain at its mark.
3. Stop when the port-lip witness reaches the mouth and the ram meets its travel stop. Do not force the lever beyond the stop.
4. If the mark and stop disagree, or the tube slips or the sleeve folds, release the load and check the depth, marks, grip position and alignment.

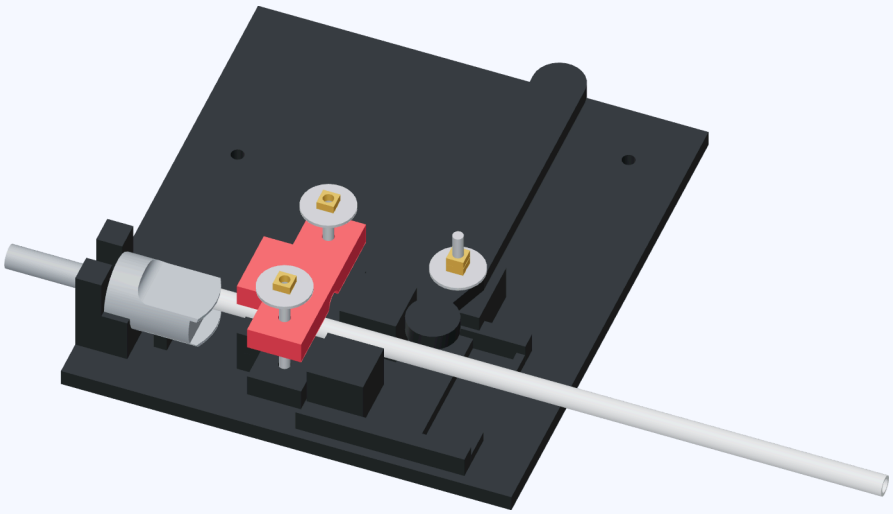
A STOP LIMITS TRAVEL

The stop is a geometric overtravel limit. It does not prove full seating or load capacity. If the witness and stop disagree, recheck the setup instead of increasing force.

AFTER INSERTION · OPERATION

Open the grip and lift away

The tool comes off without pulling the sleeved tube back out of the donor.



The grip opens after releasing the load. The tube and donor remain together. The faucet lever is omitted from this CAD view; keep it attached at unpressed rest.

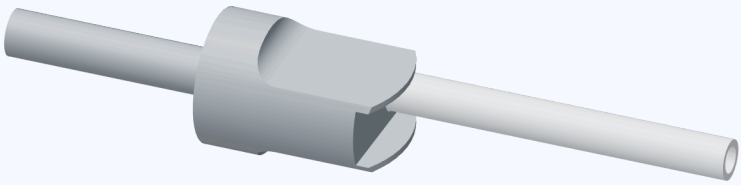
1. Move the press lever back to release the ram load. Keep the donor and tube supported.
2. Loosen the two jaw screws and open the upper jaw. Lift both tool halves off the tube without sliding a closed grip along it.
3. Lift the donor and tube together out of the nest. Leave the faucet lever attached at unpressed rest.
4. Check the seating witness and the released gripping area. The tube must remain round, without a persistent flattened section, gouge or folded sleeve.

FINISHED WHEN

The tooling is removed, the tube retains its seating witness and the released gripping area remains round.

Carry the assembly to the shell

The tube, thimble and faucet lever remain attached to the Touch-Flo. The insertion tooling comes away.



The donor and sleeved tube after removal from the jig. The faucet lever is omitted from this CAD view; keep it attached for shell entry.

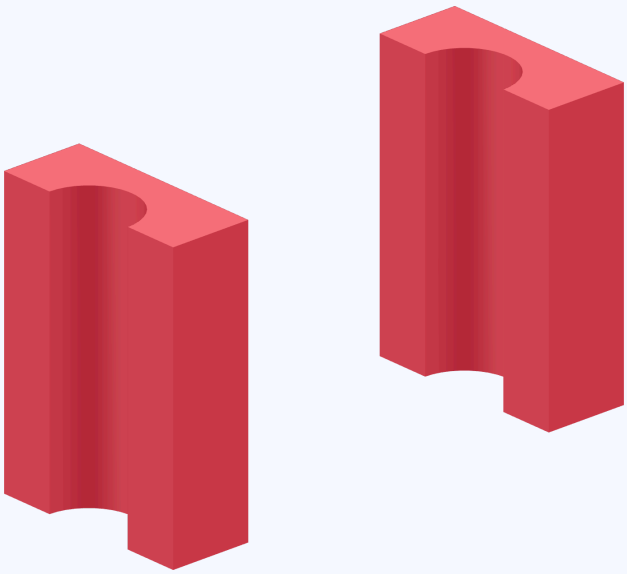
1. Follow the matching faucet assembly guide for the tube routing and shell entry.
2. Flip the attached faucet lever up to its entry position before inserting the donor into the shell.
3. As the donor seats, the shell pushes the faucet lever forward. Once fully seated, the shell prevents that flipped-up position.
4. Allow the faucet lever to snap down to its **unpressed rest** position and check its normal actuation and return.

FINISHED WHEN

The donor is seated, the tube retains its sleeve and seating witness, and the faucet lever rests down without being pressed.

Keep the gripping surfaces clean

The tool is an unprinted prototype. The assembly order comes from the observed faucet build; the fixture's grip and insertion performance remain unmeasured.



Replaceable F liners. Their smooth channels and full flat backings carry the tube load without teeth.

1. Remove strings or debris from the liners, rail faces, stop and donor nest before use. Keep the cap-support puck's hole clear.
2. Replace a torn, rolled or permanently distorted liner from the same prepared liner project. Do not add teeth or extra jaw squeeze to cure slipping.
3. Inspect the rigid jaws, ram, thrust wall and pivot for a crack or permanent distortion. Reprint a damaged part before using the press.
4. For storage, remove the grip from the tube and leave the liners unloaded. The ram lifts out once the lever is moved clear.

Geometry and native slicing establish shape and print preparation. They do not establish insertion force, tube retention, complete seating, fixture strength or endurance. The port-depth setup is performed on the actual donor during ordinary assembly.

Fixture source and ready print files
Observed faucet assembly sequence